

2

1

Continuum	a	Stimulus condition
Brightness	0.5	Point source
Smell	0.6	Heptane
Loudness	0.67	Sound pressure of 3000 Hz tone
Visual area	0.7	Projected square
Warmth	0.7	Irradiation of skin, large area
Taste	0.8	Saccharin
Vibration	0.95	Amplitude of 60 Hz on finger
Brightness	1	Point source briefly flashed
Cold	1	Metal contact on arm
Thermal pain	1	Radiant heat on skin
Visual length	1	Projected line
Duration	1.1	White-noise stimuli
Vocal effort	1.1	Vocal sound pressure
Lightness	1.2	Reflectance of gray papers
Taste	1.3	Sucrose
Warmth	1.3	Irradiation of skin, small area
Electric shock	3.5	Current through fingers

$$\Phi(I) \equiv kIa$$

Stevens' power law



Perceived



Actual



Exponent

Continuum	a	Stimulus condition
Brightness	0.5	Point source
Smell	0.6	Heptane
Loudness	0.67	Sound pressure of 3000 Hz tone
Visual area	0.7	Projected square
Warmth	0.7	Irradiation of skin, large area
Taste	0.8	Saccharin
Vibration	0.95	Amplitude of 60 Hz on finger
Brightness	1	Point source briefly flashed
Cold	1	Metal contact on arm
Thermal pain	1	Radiant heat on skin
Visual length	1	Projected line
Duration	1.1	White-noise stimuli
Vocal effort	1.1	Vocal sound pressure
Lightness	1.2	Reflectance of gray papers
Taste	1.3	Sucrose
Warmth	1.3	Irradiation of skin, small area
Electric shock	3.5	Current through fingers

Stevens' power law

$$\Phi(I) = kI^a$$

Exponent

Perceived

Actual

Reference lines can help

